



Management Science

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To cite this article:

Hongchang Wang, Benjamin Williams, Karen Xie, Wei Chen (2024) Quality Differentiation and Matching Performance in Peer-to-Peer Markets: Evidence from Airbnb Plus. Management Science 70(7):4260-4282. <https://doi.org/10.1287/mnsc.2020.03920>

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Quality Differentiation and Matching Performance in Peer-to-Peer Markets: Evidence from Airbnb Plus

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Received: December 30, 2020

Revised: June 2, 2022; March 27, 2023

Accepted: May 5, 2023

Published Online in Articles in Advance:
August 25, 2023

<https://doi.org/10.1287/mnsc.2020.03920>

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Abstract. Matching makes or breaks peer-to-peer (P2P) platforms. As the platform-based P2P markets shift from their grassroots nature toward elite offerings, will differentiating suppliers by quality increase matching performance? Can P2P markets differentiate certain suppliers without marginalizing others? We seek answers to these questions by leveraging an empirical opportunity on Airbnb, which differentiates listings that meet high quality standards from others through its Plus program in several U.S. cities. Our findings are threefold. First, we find a sizable increase in market-level matching performance after the program launch, that is, a 15.8%–16.2% increase in the number of booked nights. Second, the rise in matching performance is attributed to reduced search frictions, evident in markets with higher discovery and evaluation costs. Third, the Plus program benefits all listing tiers through reduced search frictions, yielding the most benefits to the Plus and non-Plus listings in terms of increased bookings and prices. Regular listings also receive increased bookings because of competitive prices. Our findings hold across multiple robustness checks and offer important insights for platform design and supply management in P2P markets.

History: Accepted by Hemant Bhargava, information systems.

Supplemental Material: The web appendix and data are available at <https://doi.org/10.1287/mnsc.2020.03920>.

Keywords: matching performance • quality differentiation • search frictions • peer-to-peer markets • quasi-experiment • difference-in-differences

1. Introduction

In their humble beginnings, platform-based peer-to-peer (P2P) services or the so-called sharing economy, such as Airbnb and Uber, were often advertised as cheaper alternatives to hotels and taxis, respectively. The supply side of these P2P markets largely consisted of grassroots suppliers who shared their underutilized resources with others for extra income (Einav et al. 2016). However, this picture has recently changed. The popularity of P2P markets now attracts an increasing number of consumers desiring premium products, offering opportunities for the markets to expand into higher end territory and shift away from their modest roots. For example, Uber launched UberLUX, a luxury service through which consumers can request a ride in a top-of-the-line luxury vehicle.¹ Similarly, Airbnb rolled out Airbnb Plus, a program that features listings of exceptional quality.² Often marked as premium products and verified for consistently higher quality, these offerings tend to stand out from their

regular counterparts. Such a differentiation by quality allows P2P markets to meet the changing needs of today's consumers.

Will differentiating suppliers by quality increase matching performance? This question is central to the business model of two-sided P2P markets yet lacks a clear answer in the literature. On the one hand, quality differentiation may improve matching by reducing search frictions. Manifesting in the time and effort spent by participants to identify and complete a satisfactory transaction, search frictions usually consist of discovery costs (i.e., the costs of searching to identify an option) and evaluation costs (i.e., the costs of evaluating the fit of the option) (Mortensen 2011, Liang et al. 2019). Search theory suggests that quality differentiation, along with similar mechanisms, such as reviews and reputation systems (Moreno and Terwiesch 2014, Hui et al. 2016, Lin et al. 2018), certificates (Hui et al. 2022), authentication (Bouvard and Levy 2018, Basu et al. 2019), and recommender systems (Malgonde et al.

2020), can offer helpful information to consumers in the decision-making process. The additional information could facilitate product screening and reduce search frictions, leading to more efficient matching (Stigler 1961, Leland 1979). On the other hand, quality differentiation might increase search frictions because of overcommunication and choice overloads (Chen et al. 2020, Fong 2023), leading to longer searching/waiting time (Yang et al. 2019), which may incur customer churn. The impact of quality differentiation might also be marginal if there is already sufficient information or search frictions are low (Clemons et al. 2002, Kroft and Pope 2014). In either case, matching performance may decline or remain unchanged. As P2P markets are shifting toward more and more premium offerings, understanding the impact of quality differentiation on matching performance is crucial.

This paper studies the impacts of quality differentiation on matching performance and how such impacts vary across product tiers. We leverage a unique policy change on the largest P2P home-sharing platform: Airbnb. As of January 2023, Airbnb provides access to more than 6.6 million properties across 220+ countries and regions and is widely regarded as the unicorn of P2P sharing (Isaacson 2017).³ To offer standardized, high-quality products intended for guests looking for *beautiful stays, exceptional hosts, and added peace of mind*, Airbnb launched a program called Airbnb Plus (the Plus program or the program, hereafter) to push home sharing into the mainstream hospitality industry (Ting 2018). The Plus program features a premium tier of listings that meet hotel-like standards and are “inspected and verified in person against a 100+ point checklist covering cleanliness, comfort, and design.”⁴ Once a part of the Plus program, the listings are clearly differentiated from regular ones with a “Plus” badge displayed on their profiles. The Plus program first debuted in 13 cities worldwide in February 2018 and, subsequently, in other cities in batches.

The staggered rollout of the program across different markets at different times provides us with an empirical opportunity. We examine the change of matching performance in the markets of five U.S. cities that experienced the program (i.e., Plus-affected cities)⁵ to study the impact of quality differentiation. We apply a quasi-experiment design in which we construct a comparison group of the markets mirroring those in the Plus-affected cities but that have not experienced the program. Our identification strategy is a difference-in-differences (DID) design using both the full sample and a coarsened exact matching (CEM) sample. We also deploy a plethora of alternative specifications and analyses to ensure the robustness of our findings. For example, we use relative time models to verify that the pretreatment trends of the Plus-affected and the unaffected markets are almost identical.

We obtain data from multiple sources. We first gather Airbnb data containing monthly property-level performance metrics, which are commonly used in the hospitality industry (e.g., the number of booked nights), along with property and host characteristics. To account for economic and demographic factors, we complement the Airbnb data with a rich set of control variables collected from the American Community Survey (ACS) released by the U.S. Census Bureau. The final sample, which combines the two data sources, consists of more than 236,000 Airbnb listings in 773 zip codes from 11 cities (five Plus-affected cities and six unaffected cities) across the United States from August 2017 to November 2019 (a total of 28 months).

We measure matching performance by calculating the average number of booked nights per listing in a zip code in a month. We name this metric the *number of booked nights*, which draws a parallel to the ultimate transactions in defining matching performance (Cullen and Farronato 2020). For comparison, despite being more expensive, Plus listings receive, on average, more booked nights than non-Plus listings. They also tend to be hotel-like (entire homes, business-ready, and instantly bookable) and have superior property characteristics and host services with more stringent house rules and booking policies. Overall, Plus listings are emblematic of Airbnb’s convergence from alternative accommodations to the mainstream lodging space.

Our main analysis focuses on investigating the changes in matching performance after the program launch to determine the impact of quality differentiation. If quality differentiation, instantiated by the Plus program, indeed facilitates matching, we should observe an increase in the number of booked nights in Plus-affected markets compared with unaffected markets. Our results confirm this conjecture. The number of booked nights increases by about 0.5–0.6 nights, representing a 15.8%–16.2% leap in matching performance in Plus-affected markets. This is in line with the search theory advocating that quality differentiation reduces search frictions in two-sided markets. After all, who enjoys tediously searching through an abundance of diverse options to find the perfect listing? By differentiating suppliers, platforms seem to facilitate matching effectively.

If the reduction in search frictions is indeed the underlying mechanism, the Plus program should exhibit a more prominent impact in markets with higher search frictions. In markets with higher discovery costs (more listing options) and evaluation costs (more heterogeneous listings), the Plus program should further amplify matching performance. To test this, we construct two measures as proxies for the two sources of search frictions in P2P markets: market thickness and rating dispersion. The former is measured by the number of available listings in a market, representing discovery

costs. The latter is measured by the standard deviation of the listings' review ratings in a market, representing evaluation costs. Consistent with our prediction, using market thickness and rating dispersion as moderators, we find a larger increase in matching performance in markets with more listings to discover and higher heterogeneity to evaluate. These findings support the mechanism behind the Plus program: the effect of quality differentiation is more prominent in markets with higher search frictions because of reduced discovery and evaluation costs.

Although the program seems to increase the overall matching performance in the market, does the increase come at the cost of marginalizing non-Plus listings? We answer this question by analyzing the program's impact on three groups/tiers of listings: Plus listings, close-to-Plus listings (listings similar/comparable to Plus listings but not part of the Plus program), and regular listings. Surprisingly, we find matching performance increases significantly for all three listing groups. For the Plus and close-to-Plus listing groups, the increase in booked nights directly benefits from the reduced search frictions. This is because the Plus program primarily helps consumers distinguish the Plus listings from close-to-Plus listings, which may be hard to tell without the Plus program. We further find the Plus and close-to-Plus listings charge higher nightly prices, securing increased booked nights because of the reduced search frictions. Such a price rise in these two groups introduces more booked nights to the regular listings, which see an insignificant change in price and, thus, benefit from the relative price advantage compared with the other two groups. Together, these findings do not suggest that quality differentiation marginalizes non-Plus listings; rather, the quality differentiation induced by the Plus program makes the whole market more efficient and benefits all product tiers on the supply side. Further, we find the platform is better off in market-level matching performance. Our back-of-the-envelope calculations show an increase in booked nights ranging from 85.04 to 98.60 in a typical Airbnb market.

We rule out several alternative explanations potentially confounding the impact of the Plus program (and its mechanism of reduced search frictions): changes in listing supply, changes in available nights, potential entrants and exits, changes in host services, effects of Airbnb market characteristics, alternative matching systems or web designs, changes in demand, and potential selection bias. We also check the robustness of the findings through two-way fixed effects estimators with heterogeneous treatment effects, alternative measures of matching performance, alternative measures of discovery and evaluation costs, city-level clustered standard errors, alternative forms of the dependent variables, city-specific trends and seasonality, and a placebo test based

on randomly assigned treatments. Our conclusions remain consistent in all these additional analyses. For easy reference, Table A1 of the web appendix summarizes all these tests.

Our empirical findings provide important implications for the platform-based P2P markets. As suppliers continue to grow and diversify in these markets, it is increasingly costly for consumers to identify an appropriate match. The need to discover a suitable match from a plethora of options coupled with the need to evaluate heterogeneous offerings calls for effective platform designs to mitigate search frictions. Unlike prior methods, such as consumer reviews, third-party certification/authentication, etc., the Plus program is a costly solution involving physical off-line inspection and verification by the platform. Traditional firms and industries may have used this type of quality differentiation, but it is rare and potentially risky for online platforms. Our findings demonstrate it is worth investing in quality differentiation to segment suppliers, especially in markets with a higher density of listings and a higher level of listing heterogeneity. We also show the program's differential impacts on various tiers of suppliers. Such a platform initiative benefits suppliers in all tiers by facilitating the matching between two sides of the market and allows higher quality suppliers (Plus and close-to-Plus listings) to charge more without undermining the performance of regular listings. Because quality differentiation reduces search frictions and makes the market more efficient, all groups of listings can increase matching performance. As P2P platforms continue to grow on a global scale, we provide a useful reference revealing whether and how introducing a platform-initiated quality differentiation effort such as the Plus program may affect matching performance. We consider our study as a call for future research on platform design and supplier management to continue addressing matching problems in P2P markets.

The rest of the paper is organized as follows. In Section 2, we summarize the related literature and our contributions. We describe the institutional setting in Section 3 by introducing the Plus program, the data, and our empirical strategies. The main findings and additional analyses regarding alternative explanations and robustness checks are reported in Section 4. Finally, we make concluding remarks in Section 5 and furnish a plethora of appendices supporting the main results in the web appendix.

2. Related Literature

Our study is derived in the context of two-sided markets, in which buyers and sellers transact with significant search frictions. This makes two-sided markets an ideal context for the study of matching performance.

Previous works in this space approach matching performance via either theoretically modeling the dynamic processes of matching or empirically studying the market design facilitating matching (Mortensen 2011, Lin et al. 2018, Basu et al. 2019, Ashlagi et al. 2020, Cullen and Farronato 2020). These studies range from examining matching barriers found in the markets, such as search frictions (Mortensen 2011) and authentication issues (Basu et al. 2019), to introducing strategies for improving matching via platform recommendations (Ashlagi et al. 2020) and geographic expansion (Cullen and Farronato 2020). They measure matching performance through metrics such as occupancy rates (Li and Netessine 2020), matching ratios (Lee et al. 2020), and binary matching classifications such as matched/not matched (Lin et al. 2018). The consensus of these measures is that they all rely on ultimate transactions to calculate matching performance. Our measure of matching performance, the number of booked nights, aligns well with the current literature because it examines the change in the number of transactions in a two-sided (guests and hosts) P2P accommodation market.

We complement the existing research by exploring quality differentiation as an emerging mechanism to alleviate search frictions in these markets. By focusing on the impact of quality differentiation (through the Plus program) on matching performance in one of the largest two-sided platforms, our work primarily contributes to four streams of literature: search theory and matching, search frictions, platform design, and P2P markets.

2.1. Search Theory and Matching

Our work is first related to the classic literature on search theory and matching. This literature shows that information asymmetry in markets leads to search frictions and market inefficiency (Stigler 1961, Burdett and Judd 1983, Tadelis and Zettelmeyer 2015, Mayya et al. 2021). Because two-sided markets are increasingly characterized by a high degree of heterogeneity among suppliers, there is an urgent need for effective mechanisms and market designs to maximize matching performance (Chade et al. 2017). Existing research shows information technology may reduce search frictions in general (Overby and Forman 2015). Such technology manifests in our study as a platform-initiated program (i.e., the Plus program) centering on quality differentiation among suppliers.

Quality differentiation is defined as tiering suppliers by their quality to create market segments and alleviate competition. By differentiating suppliers by quality, platforms essentially create tiers of suppliers in which the tiers represent various levels of quality (Shaked and Sutton 1982, Gabszwick and Thisse 1986). This is true in the case of the Plus program, in which Plus listings can

be considered as one listing tier that stands out from other listings. Whereas quality differentiation is first studied in industrial organizations (Moorthy 1988), recent literature focuses primarily on firm competition strategies and product offerings (Jing 2017, Jiang and Yang 2019). The ever-changing customer needs present a new matching challenge that calls for a revisiting of the efficacy of quality differentiation (Bhargava et al. 2013, Chen et al. 2020). This is especially important and controversial for online platforms, which normally stay away from costly off-line activities. By investigating whether the tiering of suppliers via quality differentiation affects matching performance in two-sided markets, our study makes a timely contribution to this stream of literature.

2.2. Search Frictions

We propose that quality differentiation drives matching performance by reducing search frictions. Search frictions are a major problem in platform-based, two-sided markets and are critical to overcome. Quality differentiation is a reasonable strategy for solving this problem. However, the effects of platform-initiated quality differentiation on matching performance are largely unknown. Li and Netessine (2020), studying a P2P accommodation platform, point out that limiting choices for consumers (i.e., reducing search frictions) can improve matching performance. Therefore, differentiating supply by a measure of quality (i.e., the Plus program) might be a valuable tool to improve matching performance because it can reduce search frictions on the Airbnb platform.

From the consumer's perspective, search frictions represent the time and effort spent finding and closing a satisfactory transaction with a supplier. In the context of two-sided markets, search frictions primarily include two types of costs: discovery and evaluation costs (Mortensen 2011, Liang et al. 2019). Discovery costs occur when consumers search for potential matches among all options (e.g., via screening, filtering, or ranking). Quality differentiation could reduce discovery costs by highlighting certain options and facilitating screening. Evaluation costs occur when consumers evaluate potential matches (e.g., reading reviews, verifying listing information, or communicating with suppliers) to make final decisions. By ensuring listing and service quality through platform-initiated inspection and certification, quality differentiation could reduce evaluation costs among heterogeneous offerings by minimizing review readings or guest–host communications. Taken together, the Plus program may work as a quality differentiation strategy that will eventually reduce both discovery and evaluation costs toward improved matching performance. We theorize and measure the discovery and evaluation costs, respectively, using two proxies: market thickness and rating dispersion.

A larger number of suppliers make a “thicker” market by increasing the inventory and the chance a specific demand is satisfied (Halaburda et al. 2018, Bimpikis et al. 2020, Li and Netessine 2020, Cullen and Farronato 2020). However, the discovery costs grow with increased options (Fradkin 2017, Arnosti et al. 2019, Horton 2019, Kanoria and Saban 2021), making it more difficult to find a match (Chen et al. 2020). There is a long-standing discussion regarding the trade-off between market thickness and discovery costs in platform growth, debating whether platforms can obtain sizeable returns by operationally thickening their markets (Li and Netessine 2020). The literature, however, has not reached conclusive findings. We add insights to this debate by empirically investigating how market thickness interacts with quality differentiation (which supposedly decreases discovery costs) in determining matching performance. Specifically, we find that, in thick markets, it is in a platform’s interest to differentiate quality among suppliers for quick discovery of the desired products/services.

Evaluation costs exist when products are heterogeneous with different levels of quality and price (Smith and Brynjolfsson 2001, Clemons et al. 2002). In two-sided markets, especially accommodation markets, products and services are more likely heterogeneous than homogenous as evaluated by various characteristics (location, property type, number of rooms, etc.). Markets with a higher degree of heterogeneity are more prone to various forms of evaluation costs (Horton 2019, Li and Netessine 2020, Cullen and Farronato 2020). Particularly when there are more products with different combinations of quality and price, consumers must spend more effort verifying the stated qualities, comparing within and between multiple segments, and evaluating their final options, leading to higher evaluation costs. We indeed find a stronger impact of quality differentiation on matching performance in markets with a higher degree of heterogeneity. Our results add new evidence to how evaluation costs in a market affect the impact of quality differentiation on matching performance.

2.3. Platform Design

We further benefit from and contribute to the platform design literature. Prior research in this stream documents various systems in place to improve platform design to reduce search frictions between buyers and sellers. These designs include warranties (Grossman 1981), buyer protection policies (Hui et al. 2022), feedback and reputation systems (Moreno and Terwiesch 2014, Tadelis 2016, Lin et al. 2018, Benson et al. 2020), and interactive filters and sorting tools (Haubl and Trifts 2000) as well as certification (Leland 1979) and authentication (Bouvard and Levy 2018, Basu et al. 2019). Although these systems generally work well, several studies argue that they may be subject to biases

such as inflated reputation (Bolton et al. 2019, Zervas et al. 2021) and self-selection of reviews (Liu and Karahanna 2017).

The focus of this study, the Plus program, is an alternative platform design evaluating whether the suppliers meet a platform-determined quality threshold. The underlying channel, quality differentiation, is also different from the well-studied approaches in prior literature. We contribute to the literature by providing a new perspective on whether quality differentiation leads to a sizeable increase in matching performance, how this effect may accelerate (or backfire) in markets with varying levels of search frictions, and the welfare implications for different tiers of suppliers. Considering that quality differentiation is likely more costly (e.g., it requires off-line, platform-involved verification and inspection) than existing platform designs, this study also offers a reference for evaluating the potential returns to quality differentiation investment for platform-based, two-sided markets attempting to differentiate suppliers based on verified quality.

2.4. P2P Markets

Our study adds to the fast-growing literature on P2P markets. There is a burgeoning field examining P2P markets fueled by the increasing use of technologies, allowing individual suppliers to capitalize on their underutilized products and services at low costs (Athey and Gans 2010, Seamans and Zhu 2014). Existing research focuses largely on two key areas.

The first area is the competition between disruptive P2P market entrants (mainly Uber and Airbnb) and incumbents offering similar goods or services. Literature in this area examines the impacts of Uber on taxi services (Cramer and Krueger 2016), public transit (Babar and Burtch 2020), private vehicle sales (Gong et al. 2023), and the impacts of Airbnb on hotels (Li and Srinivasan 2019, Zervas et al. 2021). The second area is the externality and spillover of P2P markets to other types of markets. For example, existing studies examine Uber’s effect on local entrepreneurial activities (Burtch et al. 2018) and vehicular homicide rates (Greenwood and Wattal 2017). A growing body of work investigates Airbnb’s impact on local rents and/or house prices (Horn and Merante 2017, Barron et al. 2020, Chen et al. 2022a), local restaurants (Alyakoob and Rahman 2022), the labor force in the tourism industry (Fang et al. 2016), and noise disturbance to neighborhoods (Filippas and Horton 2017).

Different from the literature in these two areas, this study shifts the focus from observing the external influence of P2P markets to examining the internal design and supplier management within P2P platforms (Li et al. 2022). Dewan et al. (2019) examine the impact of the Plus program from the same perspective. Whereas Dewan et al. (2019) focus on the spillover effects of Plus

listings on non-Plus listings, we differ by focusing on how the Plus program reduces search frictions at the market level. Specifically, we investigate if differentiating peer suppliers by quality drives matching performance on P2P platforms, and we pinpoint the underlying mechanism of reduced search frictions behind the effect. With a particular emphasis on the differential impacts of quality differentiation on three tiers of suppliers, we also highlight the importance of platform-initiated supplier management and the corresponding welfare implications for these suppliers. Our insights complement research on P2P markets by adding a less-studied perspective and a call for continuous research on the market design of P2P platforms.

3. Institutional Setting

3.1. Airbnb and the Plus Program

In Airbnb’s early days, grassroots hosts with underutilized spaces, such as a spare bedroom or basement, played a significant role in helping the platform grow and scale. These spaces were often seen as cheaper alternatives to hotel rooms. Whereas some travelers back in 2008, when Airbnb first debuted, were satisfied with the bare minimum of quality and amenities, they now expect much more, and so does Airbnb. The Plus program represents Airbnb’s belief that it needs to take home sharing into the mainstream hospitality space, that is, in which hotels reside. Whereas some see the Plus program as part of the natural progression for Airbnb as it grows and matures, others see it as a symbol of creeping commercialization, hurting the image Airbnb has cultivated for itself over the past decade (Ting 2018).

The Plus program officially launched in February 2018 in 13 cities around the world. In the United States, San Francisco was among the first batch of cities.⁶ The program quickly spread to other U.S. cities, including Denver; Nashville; New Orleans; and Washington, DC, in November 2018. Figure 1 presents a timeline of the staggered rollout of the program across these cities within our study period between August 2017 and November 2019, a total of 28 months. The study period covers six months before February 2018 (the first program rollout) and 12 months after November 2018 (the second program rollout). There were no other major platform-initiated programs such as the Plus program in these cities between 2017 and 2019.

To apply for Plus status, a host must pay a one-time \$149 application fee. After approval by Airbnb, the host receives a “Plus” badge for the host’s listing and access to in-home services such as professional photography, customized design consultations, and premium support from Airbnb. These program offerings provided by Airbnb borrow heavily from the way hotel brands manage the relationship with their hotel owners and franchisees (a possible inspiration for the Plus program). Symbolically, the addition of the Plus program to the Airbnb platform represents a maturation of the company’s product mix—a sign that Airbnb is no longer simply a cheaper alternative to a hotel and that, in some ways, it is beginning to resemble the hotel industry.

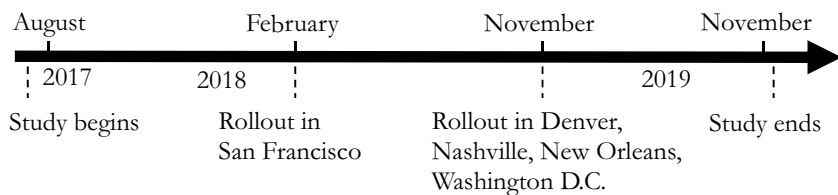
To qualify for the Plus status, the listing must maintain an average rating of 4.8/5 without canceled reservations over the past year. Next, each listing is verified first by a remote visual inspection and then via an in-person assessment. The assessment is performed by a third party, relying on a quality checklist of 100+ points applying to not only the physical characteristics of the listing (e.g., from fast wi-fi and streaming-ready televisions to kitchens with all the cooking essentials), but also the service provided by its host (e.g., house rules, check-in guide).⁷ Afterward, the host must remedy any issues raised by the inspector to meet the program standards. These Plus listings are not necessarily entire places or considered luxury; private rooms can join the Plus program as long as they meet the requirements.⁸ This is similar to hotels that are not always luxury experiences but rather stay meeting certain standards and qualifications (e.g., bed with fresh sheets, bathroom stocked with soap/shampoo, television, etc.). An Airbnb listing may receive Plus status if it is merely a small room in a house or if it is a luxurious villa, so long as it meets the quality standards.

3.2. Data and Measures

The Plus program, by differentiating quality among listings, provides a unique opportunity to examine the impact of quality differentiation on matching performance in the P2P marketplace, which is the research interest of our study. Data pertaining to our study are from two sources: Airbnb and the U.S. Census Bureau.

We first obtained Airbnb data through an independent, noncommercial research platform that tracks Airbnb properties over time in cities around the world.

Figure 1. Timeline of the Plus Program Rollout Across Sampled U.S. Cities



Its data service is widely trusted and used by policy-makers (e.g., the city of Los Angeles) and academics (e.g., Gurran and Phibbs 2017, Wachsmuth and Weisler 2018). We follow prior literature (Li and Srinivasan 2019, Chen et al. 2022b) to define active listings as those marked as either reserved or available on a host's calendar. For each property in a year-month, we have rich information, including the listing characteristics (e.g., whether a listing is business-ready; is instantly bookable; is an entire place, shared room, or private room; has a cancellation policy along with the number of guest reviews and the review ratings for the listing), the host characteristics (e.g., whether a host is a super host and the response rate to inquiries), and the transaction performance (e.g., nightly price and the number of booked nights).

We complement the Airbnb data with the ACS data, released annually by the U.S. Census Bureau. The annual census data contains information on population, number of housing units, employment rate, household income, median rent, number of renter-occupied housing units, age of construction, level of education, etc., in local markets (zip codes). We control for these variables in our analysis because the hospitality industry (of which Airbnb is a part) is not immune to local socioeconomic and demographic characteristics.

We merge these two data sources at the zip code level by year-month, which is the unit of analysis in our empirical specifications. We focus on the zip code level rather than the city level because the former provides more precise and relevant information about the accommodation market. Considering consumers' search behavior of accommodations, a market at the zip code level is appropriate because they often know where they plan to go and pin down one or two areas in a city rather than the whole city in search. Our observation period is from August 2017 to November 2019 (a total of 28 months). During our study period, there were more than 236,000 Airbnb listings in 773 zip codes of the sampled cities. The data offer several advantages for studying the impact of the Plus program. First, the richness of the data allows us to observe individual listing-level performance metrics (number of booked nights, nightly price, etc.) controlling for any time-varying covariates. Second, the data contain many geographically dispersed markets, allowing us to associate specific market contexts, such as market thickness and rating dispersion in a market as well as local socioeconomic and demographic characteristics with matching performance. Finally, our study period covers a sufficiently large time window for the program's impact to unfold and manifest.

Table 1 presents the definitions of major variables in this study. Our dependent variable is the average number of booked nights per listing in a zip code in a

month. Our primary interest is to estimate the impact of *AfterPlus* (the program dummy), controlling socioeconomic and demographic characteristics. The moderators of market thickness and rating dispersion allow us to unveil the mechanism of the program's impact with respect to discovery and evaluation costs, respectively. In the analyses, we take the logarithm of several socioeconomic and demographic characteristics (i.e., the control variables), which exhibit skewed distributions. For consistency with other variables, we report their raw values in this table for easy comprehension.

3.3. Plus Listings on Airbnb

The premise of the Plus program centers on the platform's ability to differentiate the quality of heterogeneous listings into higher quality listings (the Plus listings) and non-Plus ones. When applying to the Plus program, the hosts must meet a strict checklist and service standards (i.e., 100+ checkpoints) to prepare their properties to be hotel-like standardized, higher quality listings. To check if this premise holds, we compare the characteristics of Plus and non-Plus listings. To conduct a fair comparison, we use characteristics measured prior to the program launch. We examine whether listings identified as Plus (after the program launch) were indeed of higher quality than non-Plus ones.⁹

Table 2 reports the summary statistics on characteristics of listings and hosts along with listing performance (number of booked nights, nightly price, etc.), of Plus and non-Plus listings. Plus listings indeed exhibit a higher level of quality than non-Plus listings across several aspects. First, on average, Plus listings receive more guest reviews and higher review ratings than non-Plus listings. They also have more bedrooms, bathrooms, and guests allowed and are more hotel-like (entire homes, business-ready, and instant bookable). Along with higher quality, Plus listings have more stringent house rules, including a stricter cancellation policy, longer minimal nights of stay, and larger financial commitment of guests (e.g., higher security deposit, cleaning fees, and fees for additional guests). Second, hosts of Plus listings tend to provide better services than those of non-Plus listings as they are more likely to respond to guest inquiries. Super hosts¹⁰ are also more prevalent among hosts of Plus listings. Finally, potentially because of the higher quality of properties and services, Plus listings charge higher prices than non-Plus listings with a significant difference of about \$67 per night (\$259.12–\$191.98). In spite of the higher average nightly price, Plus listings receive more bookings per month. Overall, our findings confirm the effectiveness of the Plus program in differentiating listings: listings in the Plus program have higher quality, better service, and superior performance.

Table 1. Variable Definitions and Summary Statistics (Zip Code by Year-Month)

Variable	Definition	Mean	Standard deviation	Minimum	Maximum
Matching performance					
Number of booked nights	The average number of booked nights of the next month for all listings in a zip code in a month (a negative value implies more cancellations than reservations) ^a	3.353	2.980	−6.0	15.0
The Plus program					
AfterPlus	Dummy variable indicating whether the Plus program was rolled out, equal to one for zip codes in San Francisco after February 2018 and zip codes in Denver; Nashville; New Orleans; and Washington, DC after November 2018	0.152	0.359	0	1
Search frictions					
Market thickness	The average number of listings in a zip code prior to the Plus program	160.15	285.58	1	2,376.8
Rating dispersion	The average standard deviation of the overall review rating ^b in a zip code prior to the Plus program	6.652	3.954	0	39.12
Socioeconomic and demographic characteristics (ACS controls)					
Population	Population	32,077	23,664	20	114,647
Percentage employment	Percentage of the employed population	0.946	0.030	0.55	1
Median income	Median income of households	77,765	33,654	15,000	250,001
Median rent	Median rent of households	6,733	6,702	5	32,785
Number of housing units	Number of housing units	13,436	9,467	11	48,734
Percentage renter-occupied units	Percentage of renter-occupied housing units	0.438	0.224	0	1
Percentage income spent on rents	Percentage of the median rent relative to the median household income	0.22	0.25	0.01	0.81
Percentage higher education	Percentage of the population with a high school degree or higher	0.66	0.29	0	1
Median construction age	Median age of all constructions	54	21	7	80

^aA visualization of how the number of booked nights was calculated is reported in Web Figures A1.1 and A1.2. Because guests usually book for future stays, we use a formula to calculate the number of booked nights of the next month for a listing that occurs in the current month: booked nights occurring in the current month = available nights of the next month observed in the current month − available nights of the same month observed in the next month. This is possible because we can observe availability for a listing several months into the future. For example, suppose that on July 1, 2017, we observed 25 available nights for a given listing in August 2017. If there are 25 available nights for August 2017 as observed on July 1, 2017, and there are 18 available nights for August 2017 as observed on August 1, 2017, then the number of booked nights of the next month for this listing in the current month is 25 − 18 = 7 nights. To measure matching performance, we average the number of booked nights for all listings in a zip code to determine the average number of booked nights per listing in a zip code.

^bConsistent with the literature (e.g., Chevalier and Mayzlin 2006, Zhu and Zhang 2010, Chen et al. 2022b), we used the overall review rating of each listing when calculating rating dispersion. The overall review rating is measured on a 0–100 scale, and 100 equals five stars.

Table 2. Comparing Plus Listings with Non-Plus Listings

	Plus listings				Non-Plus listings				<i>t</i> -statistic	<i>p</i> -value
	Mean	Standard deviation	Minimum	Maximum	Mean	Standard deviation	Minimum	Maximum		
Listing characteristics										
Number of reviews	52.98	54.67	0	389	33.93	54.31	0	639	15.57	<0.01
Review score rating	98.38	2.10	80	100	93.91	7.94	20	100	79.91	<0.01
Number of bedrooms	1.69	1.04	0	6	1.52	1.14	0	15	7.2	<0.01
Number of bathrooms	1.46	0.71	0	5.5	1.36	0.87	0	15	6.01	<0.01
Number of guests accommodated	2.38	1.94	1	12	1.95	1.73	1	34	9.96	<0.01
Entire home (0/1)	0.81	0.39	0	1	0.66	0.48	0	1	17.91	<0.01
Business ready (0/1)	0.54	0.50	0	1	0.18	0.39	0	1	28.86	<0.01
Instantly bookable (0/1)	0.44	0.50	0	1	0.35	0.48	0	1	8.55	<0.01
Flexible cancellation (0/1)	0.06	0.24	0	1	0.22	0.41	0	1	−27.45	<0.01
Minimum nights of stay	3.91	6.75	1	31	3.65	8.06	1	359	1.52	0.13
Security deposit	304.41	215.19	0	950	240.36	191.78	0	999	9.95	<0.01
Cleaning fee	99.20	57.46	0	450	79.86	59.63	0	850	13.02	<0.01
Fee for additional guest	23.78	36.36	0	300	16.04	27.70	0	300	9.56	<0.01
Host characteristics										
Host response rate	99.67	3.43	0	100	97.36	10.71	0	100	23.96	<0.01
Super host (0/1)	0.67	0.47	0	1	0.30	0.46	0	1	30.76	<0.01
Listing performance										
Nightly price	259.12	182.17	0	990	191.98	150.01	0	999	14.41	<0.01
Number of booked nights	4.85	7.03	−25	30	3.39	7.38	−30	30	6.97	<0.01

3.4. Empirical Strategies

We show that the Plus program effectively differentiates listings by quality. We now investigate how this quality differentiation program affects the matching performance of Airbnb markets. The fact that the program launched across markets at different times provides us with a quasi-experimental opportunity to examine its effect. We apply a DID strategy capitalizing on the staggered rollout of the Plus program in U.S. cities. We focus on markets (zip codes) in five U.S. cities affected by the Plus program, including San Francisco; Denver; Nashville; New Orleans; and Washington, DC zip codes in these cities serve as the affected group (i.e., Plus-affected) whereas zip codes in other comparable but unaffected cities serve as the unaffected group (i.e., Plus-unaffected).

To find the comparison cities, we first removed those without data before November 2018 (the date of the program launch for the second batch of cities). Our data source yielded eight cities/regions. Further, we dropped regions (San Mateo County) with too few listings (seven listings in total) and cities (Jersey City) with close proximity to and possible data overlap with New York City (a city in the control group for which we had data). The search yielded six comparable cities/regions available in our data: New York City, Las Vegas (Clark County), the greater Boston area (Boston, Cambridge, and Salem), and the Twin Cities. The full sample consists of 773 zip codes with 161 zip codes in the five affected cities and 612 zip codes in the six unaffected cities/regions.

For a zip code i in year-month t , we model the matching performance in this market as a function of the Plus program along with other control variables. The equation is as follows:

$$Y_{it} = \beta \cdot \text{AfterPlus}_{it} + \gamma \cdot X_{it} + u_i + v_t + \varepsilon_{it}, \quad (1)$$

where the matching performance, Y_{it} , is measured by the number of booked nights for listings at zip code i in year-month t .¹¹ The term AfterPlus_{it} represents whether the Plus program was launched at zip code i in year-month t . For any zip code in the affected group, the value of this term is one in and after the month of program launch and zero otherwise. For any zip code in the unaffected group, the value of AfterPlus_{it} is always zero.¹² Its estimated coefficient, β , captures the impact of the Plus program on matching performance. We use X_{it} to represent a vector of ACS controls (as in Table 1). We also include two-way fixed effects, u_i and v_t , at the zip code and year-month level as in standard DID designs. The residual term, ε_{it} , captures idiosyncratic shocks.

One concern of the DID specification is that the zip codes in the unaffected group may not be good counterfactuals for the affected zip codes. For stronger comparability, we construct another sample using a CEM

approach (Blackwell et al. 2009, Iacus et al. 2012). Specifically, we match Plus-affected zip codes with those unaffected zip codes based on the mean value of observed local socioeconomic and demographic characteristics (population, number of housing units, etc.) as well as Airbnb market characteristics (total listings, number of booked nights, etc.) up until the point Plus program was launched.¹³ A by-product of this matching process is that some unobserved variables not used in the CEM also achieve a good level of balance, indicating that the CEM approach alleviates imbalance. That is, if any unobserved variables contribute to the program launch, the CEM largely addresses the selection concern by making the affected group and the unaffected group comparable before the treatment (Wang and Overby 2021).

As a result of the CEM process, we identify 438 zip codes in the unaffected group, best mirroring the 148 zip codes in the affected group for a total of 586 matched zip codes.¹⁴ These 586 zip codes cover about 71.8% of the zip codes in the full sample. Table 3 presents the comparability of the affected and unaffected markets (zip codes) in the full and CEM samples based on the mean values of all observed characteristics up until when the Plus program was launched. The t -statistics suggest the CEM sample is more balanced than the full sample because of the insignificant difference in every characteristic between the affected markets and unaffected markets. Using both samples, we estimate the impact of the Plus program and cross-validate the results.

4. Results and Findings

As a quick road map for our results and findings, we begin by estimating the Plus program's impact on matching performance, supplemented with a relative time model to check the parallel trends assumption in Section 4.1. In Section 4.2, we explore evidence that reduced search frictions (specifically, discovery and evaluation costs) are the underlying mechanism behind the impact of quality differentiation. In Section 4.3, we further explore the program's differential impacts on various groups of listings in a market and the implications for the platform. We rule out alternative explanations in Section 4.4 and conduct robustness checks to validate our identification strategy in Section 4.5. In most analyses, we present the estimates from the DID models using both the full sample and the CEM sample to check the consistency of our findings.

4.1. Impacts of the Plus Program on Matching Performance

Our primary analyses explore whether quality differentiation—instantiated on the platform-initiated Plus program—improves the matching between sellers

Table 3. Comparability Between Affected and Unaffected Zip Codes

	Full sample (773 zip codes)				CEM sample (586 zip codes)			
	Affected mean		Unaffected mean		Affected mean		Unaffected mean	
	161 zip codes	612 zip codes	612 zip codes	Difference	148 zip codes	438 zip codes	438 zip codes	p-value
Socioeconomic and demographic characteristics								
Population (logged)	9.842	9.658	9.658	0.184	9.971	9.941	9.941	0.027
Percentage employment	0.948	0.948	0.948	−0.000	0.948	0.950	0.950	−0.002
Median income (logged)	11.17	11.18	11.18	−0.014	11.183	11.197	11.197	0.014
Median rent (logged)	8.138	7.626	7.626	0.512	8.335	8.206	8.206	0.129
Number of housing units (logged)	8.996	8.812	8.812	0.184	9.199	9.098	9.098	0.101
Percentage renter-occupied units	0.465	0.398	0.398	0.067	0.467	0.456	0.456	0.011
Percentage income spent on rents	0.181	0.226	0.226	−0.045	0.182	0.219	0.219	−0.037
Percentage higher education	0.700	0.657	0.657	0.043	0.702	0.724	0.724	−0.022
Median construction age	53.83	50.58	50.58	3.25	54.333	52.245	52.245	−1.1
Airbnb market characteristics								
Number of listings	203.33	136.87	136.87	66.46	214.78	176.88	176.88	37.90
Number of booked nights	3.616	3.885	3.885	−0.269	3.716	3.772	3.772	−0.056
Average listing price	161.15	178.90	178.90	−17.75	163.64	176.57	176.57	−12.93
								0.248
								0.830
								0.240

(hosts) and buyers (guests). By promoting high-quality listings as premium products, the Plus program may make the search more efficient because guests can more easily navigate into the product segment they prefer (e.g., Plus or non-Plus listings). If this holds true, after the program launch, we expect to see an increase in booked nights in the market-level matching performance.

To test this conjecture, we estimate the program's impact on the number of booked nights per the DID specification defined in Equation (1). We use two samples, the full and CEM samples, to compare the results. Table 4 reports the results. Column (1) presents the estimated impact using the full sample. We find that, after the program launch, the number of booked nights in an affected zip code increased by 0.531 nights. This is equivalent to a 15.8% increase ($0.158 = 0.531/3.353$, in which 3.353 is the mean number of booked nights in the full sample as in Table 1). Considering there are, on average, 160.15 listings in a zip code, the Plus program led to a total of 85.04 ($85.04 = 0.531 \times 160.15$) more nights booked in a month in the affected zip codes.

Column (2) reports the estimated impact using the CEM sample. The coefficient of 0.541 suggests an average increase of 16.2% in the number of booked nights in the affected zip codes ($0.162 = 0.541/3.333$, in which 3.333 is the mean number of booked nights in the CEM sample). The results in column (2), as do those in column (1), indicate an increase of 98.60 ($98.60 = 0.541 \times 182.26$) more nights booked in a month in the affected zip codes. The program's impact in the CEM sample is slightly greater than in the full sample because of more precise matching, but the results remain consistent across the samples. Together, the results suggest that market-level matching performance benefits significantly from quality differentiation.

A prerequisite for a DID estimation is the exogeneity of the treatment (the program launch in our case), which usually manifests in the pretreatment parallel trends assumption. Ideally, the affected and unaffected zip codes should have parallel trends before the program launch. The subsequent change in matching performance may then be attributed to the Plus program. We adopt the relative time model (Autor 2003, Angrist and Pischke 2008) to check if similar pretreatment trends exist for the affected and unaffected zip codes. The equation is

$$Y_{it} = \sum_{\tau=-4}^{-2} \lambda_{\tau} D_{it+\tau} + \sum_{\tau=0}^4 \lambda_{\tau} D_{it+\tau} + \gamma \cdot X_{it} + u_i + v_t + \varepsilon_{it}. \quad (2)$$

Essentially, we replace the program dummy, $AfterPlus_{it}$, in Equation (1) with relative time dummies, $D_{it+\tau}$, in Equation (2) for the affected zip codes. Note that these dummy variables are always equal to zeros for

Table 4. Impacts of the Plus Program on Matching Performance

Dependent variable: Number of booked nights	(1) Full sample	(2) CEM sample
AfterPlus	0.531*** (0.123)	0.541*** (0.149)
ACS controls (socioeconomic and demographics characteristics)		
Population (logged)	1.762 (2.668)	6.414* (3.794)
Percentage employment	−13.539** (5.241)	−6.111 (6.883)
Median income (logged)	1.322 (1.044)	1.591 (1.151)
Median rent (logged)	0.982 (0.952)	3.805 (2.716)
Number of housing units (logged)	−0.207 (2.740)	0.711 (5.243)
Percentage renter-occupied units	−4.229 (4.270)	−5.974 (8.194)
Percentage income spent on rents	2.172*** (0.652)	3.781** (1.619)
Percentage higher education	−0.507 (0.331)	−1.236* (0.700)
Median construction age	0.021 (0.029)	0.093** (0.046)
Zip code fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Observations	13,176	10,195
Adjusted R ²	0.215	0.219

Notes. For brevity, we omit the estimates of ACS control variables in the following tables. Robust standard errors clustered at the zip code level are shown in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

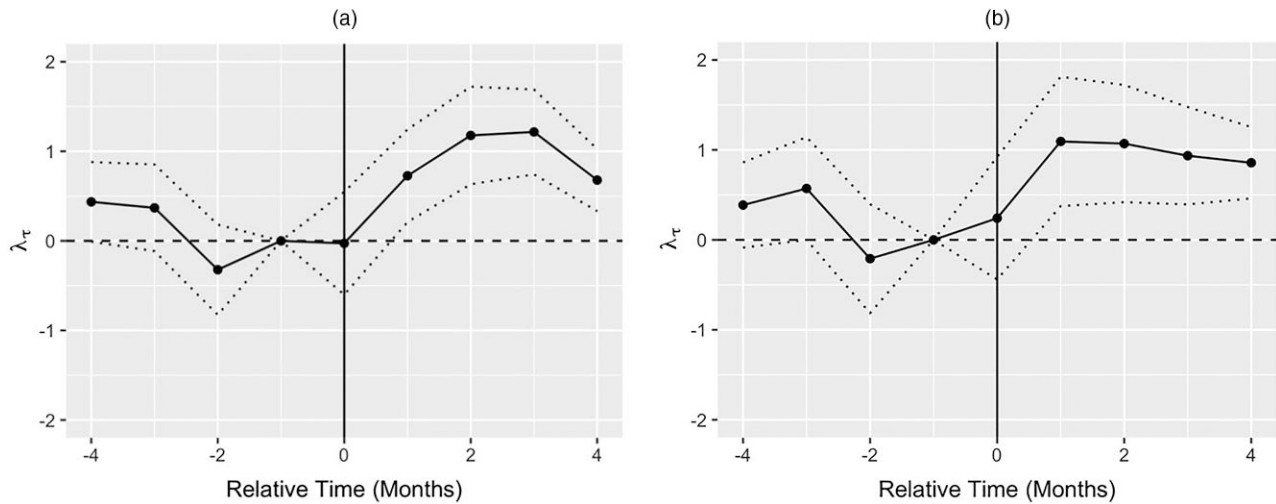
the unaffected zip codes. The set of coefficients λ_τ identify whether similar pretreatment trends exist in the affected and unaffected zip codes and how the program’s impact evolves over time at each phase. We estimate Equation (2) with time periods over four months away from the aggregated program launch to the -4 (or $+4$) dummies. We assume the status of matching performance beyond four months before the treatment does not influence or carry over to the current market. We also assume matching performance becomes stable beyond four months posttreatment. Because the program’s impact might be immediate upon its launch, we use the -1 relative time dummy as our baseline to check the parallel trends.

We estimate the relative time model using two samples and report the full sample results in column (1) and the CEM sample results in column (2) of Table 5. The estimates in both columns suggest an insignificant change in the difference in the number of booked nights between affected and unaffected zip codes before the program launch. These results support the assumption of parallel trends and indicate that the Plus program took effect after being launched in the affected zip codes. Specifically, the effects occurred almost immediately in the second month after the program launch (i.e., $+1$ relative time) and remained positive and significant in the subsequent periods. Figure 2 further shows a visual presentation of the relative time model results from Table 5 with λ_τ (the

Table 5. Relative Time Models on the Impacts of the Plus Program

Dependent variable: Number of booked nights	(1) Full sample	(2) CEM sample
Relative time ≤ -4	0.298 (0.264)	0.399 (0.297)
Relative time -3	0.322 (0.303)	0.557 (0.346)
Relative time -2	−0.369 (0.308)	−0.302 (0.352)
Relative time -1	The baseline	
Relative time 0		0.203 (0.402)
Relative time $+1$		1.128*** (0.412)
Relative time $+2$		0.898** (0.392)
Relative time $+3$		0.983*** (0.330)
Relative time $\geq +4$	0.623*** (0.215)	0.732*** (0.242)
ACS controls	Yes	Yes
Zip code fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Observations	13,176	10,195
Adjusted R ²	0.216	0.220

Notes. For brevity, we omit the coefficients of ACS controls hereafter. The ACS controls are always included in the model specification unless otherwise specified. Robust standard errors clustered at the zip code level are shown in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Figure 2. Graphic Evidence of the Parallel Trends

Notes. (a) The full sample. (b) The CEM sample.

coefficients of the relative time dummies) on the y -axis and the relative time dummies on the x -axis. The 90% confidence intervals for the coefficients are shown with dotted lines.

The graphic evidence provides strong support for the pretreatment parallel trends and the posttreatment difference between the affected and unaffected groups. Specifically, in both samples, the coefficients become significantly larger than zero after a relative time of one month. Before the program launch, the coefficients are not significantly different from zero. These results support the assumption of parallel trends and indicate the Plus program took effect after being launched in the affected zip codes.

4.2. Underlying Mechanism Behind the Program's Impacts

We show that the Plus program represents a quality differentiation opportunity to increase matching performance. If the reduced search frictions (because of quality differentiation) are the underlying mechanism behind the impact, one natural conjecture is that the Plus program exhibits a stronger effect in markets with higher search frictions. To this end, we create proxy measures of search frictions from two aspects, discovery and evaluation costs, and investigate the effectiveness of quality differentiation in markets with different levels of search frictions. If the conjecture is confirmed, we can extend the mechanism of reduced search frictions to explain the impact of the Plus program.

4.2.1. Evidence from the Discovery Costs. Discovery costs are one major source of search frictions (Fradkin 2017, Arnosti et al. 2019, Horton 2019, Kanoria and Saban 2021). We approximate discovery costs by calculating

the number of Airbnb listings in a zip code (i.e., market thickness). Market thickness is a proper proxy of discovery costs (and, thus, search frictions) because markets with more listings require guests to spend more discovery efforts even before they can identify potential matches. For each zip code, we calculate the average number of total listings in the zip code across all months before the program launch.¹⁵ We then classify zip codes into those with a higher market thickness (*Higher Market Thickness_i*) and those with a lower market thickness (*Lower Market Thickness_i*) depending on whether the zip code has more or fewer listings than the median number of listings among all zip codes in the sample (i.e., 57 listings).¹⁶

Existing literature shows a trade-off between search frictions and available product offerings in a market: thicker markets provide more options for possible matches but also increase search frictions, a hurdle to finding a match (Burdett et al. 2001, Shimer and Smith 2001, Li and Netessine 2020). Therefore, if reduced search frictions are the underlying mechanism, the Plus program should have a larger impact in thicker markets than in thinner ones because search frictions are more severe in thicker markets. We test this conjecture using a difference-in-difference-in-differences framework. Following the literature (Goldfarb and Tucker 2011, Wing et al. 2018), we add an interaction term between the program dummy and the two levels of market thickness, respectively, into Equation (1).¹⁷ Additionally, we interact the levels of market thickness with ACS controls because matching performance may be influenced differently by these covariates in thicker and thinner markets. We also split the general year-month fixed effects into two groups of year-month fixed effects based on the two levels of market thickness

Table 6. Impacts of the Plus Program by the Level of Discovery Costs

Dependent variable: Number of booked nights	(1) Full sample	(2) CEM sample
AfterPlus × Lower market thickness	0.206 (0.259)	0.063 (0.333)
AfterPlus × Higher market thickness	0.853*** (0.095)	0.892*** (0.096)
ACS controls × Lower market thickness	Yes	Yes
ACS controls × Higher market thickness	Yes	Yes
Zip code fixed effects	Yes	Yes
Year-month fixed effects × Lower market thickness	Yes	Yes
Year-month fixed effects × Higher market thickness	Yes	Yes
Observations	13,176	10,195
Adjusted R ²	0.217	0.233

Note. Robust standard errors clustered at the zip code level are shown in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

because they may not necessarily follow the same trend across time in terms of matching performance.¹⁸ The resulting model is

$$\begin{aligned} Y_{it} = & \beta_1 \cdot \text{AfterPlus}_{it} \times \text{Lower Market Thickness}_i \\ & + \beta_2 \cdot \text{AfterPlus}_{it} \times \text{Higher Market Thickness}_i \\ & + \gamma \cdot X_{it} + u_i + v_t + \varepsilon_{it}. \end{aligned} \tag{3}$$

We focus on estimating the two coefficients β_1 and β_2 in Equation (3). The former estimates the impact of the Plus program on matching performance in thinner markets, whereas the latter estimates the impact in thicker markets. By comparing the two coefficients β_1 and β_2 , we can observe how the program’s impacts differ by the level of discovery costs. Note the vector X_{it} includes the ACS controls as well as the interaction terms with two levels of market thickness.

Table 6 presents the estimation results. Columns (1) and (2) utilize the full and CEM samples, respectively. The Plus program did not have a significant impact on matching performance in markets with lower market thickness. This confirms our conjecture that the reduction of search frictions via quality differentiation is not prominent in thinner markets. This is reasonable because search frictions in thinner markets are relatively lower, and quality differentiation does not practically increase the odds that a listing is discovered.

In contrast, the Plus program has a positive and significant impact in thicker markets. The differences between the impacts in thinner and thicker markets are also significant. We conducted a postestimation test of such differences in the coefficients and obtained a p -value of 0.019 in the full sample and 0.017 in the CEM sample, showing the two coefficients are significantly different from each other. These results further support that the mechanism of reduced search frictions is at play. With more listings in a market, consumers face higher search frictions to discover a match from an abundance of options. This finding suggests that one of

the roles of quality differentiation is to facilitate matching by reducing discovery costs.

4.2.2. Evidence from the Evaluation Costs. We show quality differentiation facilitates matching in markets with higher discovery costs. To understand the mechanism behind the program’s impact comprehensively, we further examine evaluation cost and approximate it by the dispersion of consumer review ratings on listings in the market (i.e., rating dispersion). We focus on review ratings rather than other metrics because consumer reviews can reflect the quality of products and affect sales (Chevalier and Mayzlin 2006, Zhu and Zhang 2010). The review rating is a number between 0 and 100 given by Airbnb guests who complete a stay. The higher the rating, the better a guest views the quality of the listing and the associated service. We define the rating dispersion of listings in a zip code as the standard deviation of the ratings across all listings prior to the program launch. Higher levels of rating dispersion indicate a more heterogenous market in terms of quality. In markets with higher levels of rating dispersion, guests must spend more time comparing various listing options (Li and Netessine 2020). Therefore, a higher level of rating dispersion is associated with higher evaluation costs (and, thus, higher search frictions) (Hitsch et al. 2010). We classify zip codes into those of higher rating dispersion (*Higher Rating Dispersion_i*) and those of lower rating dispersion (*Lower Rating Dispersion_i*) based on whether the market has a rating dispersion above or below, respectively, the median of rating dispersion among all zip codes in the sample (i.e., 6.4).

The platform-endorsed Plus badge helps guests quickly identify higher quality listings without spending additional evaluation effort (e.g., reading lengthy guest reviews to gauge quality). By meeting the extensive quality checklist and incurring the platform’s verification cost in advance, listings with a Plus badge provide guests with the peace of mind that their quality is backed up by a platform endorsement. Considering Airbnb stays are largely experience goods of which

Table 7. Impacts of the Plus Program by the Level of Evaluation Costs

Dependent variable: Number of booked nights	(1) Full sample	(2) CEM sample
AfterPlus × Lower rating dispersion	0.290* (0.161)	0.536*** (0.164)
AfterPlus × Higher rating dispersion	1.022*** (0.197)	1.151*** (0.197)
ACS controls × Lower rating dispersion	Yes	Yes
ACS controls × Higher rating dispersion	Yes	Yes
Zip code fixed effects	Yes	Yes
Year-month fixed effects × Lower rating dispersion	Yes	Yes
Year-month fixed effects × Higher rating dispersion	Yes	Yes
Observations	11,948	9,372
Adjusted R^2	0.222	0.259

Notes. The number of observations changes because the measurement of rating dispersion is invalid in zip codes with only one listing. Robust standard errors clustered at the zip code level are shown in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

quality remains unknown until purchase, the evaluation process in reference to the Plus badge is simplified. Guests can easily seek or avoid Plus listings at a low cost. Following this reasoning, the Plus program should have a larger impact in markets with higher rating dispersion (higher evaluation costs) and a limited impact in markets with lower rating dispersion (lower evaluation costs).

To test this hypothesis, we hold everything the same in Equation (3) except for replacing the moderators with the two levels of rating dispersion.¹⁹ The resulting model is as follows:

$$\begin{aligned}
 Y_{it} = & \beta_1 \cdot \text{AfterPlus}_{it} \times \text{Lower Rating Dispersion}_i \\
 & + \beta_2 \cdot \text{AfterPlus}_{it} \times \text{Higher Rating Dispersion}_i \\
 & + \gamma \cdot X_{it} + u_i + v_t + \varepsilon_{it}.
 \end{aligned} \quad (4)$$

Table 7 presents the estimation results using the full sample in column (1) and the CEM sample in column (2). We find the Plus program is associated with higher matching performance for markets with higher rating dispersion (more heterogeneous listing offerings). The differences between the coefficients for lower and higher rating dispersion are also significant. A postestimation test of such differences obtained a p -value of 0.004 in the full sample and 0.017 in the CEM sample. This confirms our conjecture about a more prominent role of quality differentiation in facilitating matching performance in the presence of higher evaluation costs. In markets with lower rating dispersion, the impact of the Plus program is significant but smaller in magnitude. The consistent results between Equations (3) and (4) demonstrate the mechanism of reduced search frictions behind the impact of quality differentiation.

4.3. Implications for Listings and the Platform: Evidence from Three-Group Analysis

Thus far, our empirical evidence suggests the reduction in search frictions resulting from the Plus program

improves matching performance. Such an effect is more prominent in markets with more listing supplies and more heterogeneous listings. A natural follow-up question is whether the overall improved matching performance holds across the board or comes at a cost to some listings. To answer this question and inform platform design on tiering, we dig deeper into three distinct and mutually exclusive listing groups/tiers and examine the effects of the Plus program at a finer level.

The first group contains all Plus listings in a zip code. The second group contains close-to-Plus listings, which are similar/comparable to the Plus listings on listing characteristics and service quality but lack the Plus badge. The third group contains regular listings, which are the remaining listings not classified into the first two groups, and they tend to have lower quality. Arguably, the Plus program may help separate Plus from close-to-Plus listings for which the differences may be hard to tell without the program. However, the Plus program may not strongly affect the regular group through reduced search frictions; consumers who prefer regular listings can rely on, for example, nightly price or review ratings to identify these economy offerings. We aim to understand how the Plus program influences the matching performance of each group and what the potential mechanisms could be. We expect the impact to be prominent for the Plus and close-to-Plus groups but much less for the regular group.

To classify all the listings into the three groups, we implement a listing-level matching approach. Specifically, we match the Plus to non-Plus listings from both the affected and unaffected zip codes on the listing level before the program launch. The matching utilizes a rich set of listing and host characteristics, including listing prices, booked nights, review ratings, minimum nights of stay, number of bedrooms and bathrooms, number of guests accommodated, cancellation policy, super host status of the host, and whether the listing is instantly bookable. We utilize propensity score matching (PSM)

Table 8. Impacts of the Plus Program on Matching Performance of Three Listing Groups

Dependent variable: Number of booked nights	Plus listings		Close-to-Plus listings		Regular listings	
	(1) Full sample	(2) CEM sample	(3) Full sample	(4) CEM sample	(5) Full sample	(6) CEM sample
AfterPlus	0.878*** (0.168)	0.981*** (0.186)	0.521*** (0.183)	0.620*** (0.203)	0.596*** (0.117)	0.654*** (0.120)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,952	10,045	12,952	10,045	12,952	10,045
Adjusted R ²	0.306	0.291	0.292	0.270	0.240	0.288

Note. Robust standard errors clustered at the zip code level are shown in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

in this task and label the matched non-Plus listings in our PSM process as “close-to-Plus” and the unmatched/remaining listings as “regular.” This PSM process generates three groups of listings (including Plus as a group). The matched non-Plus listings are in the close-to-Plus group, and the remainder is in the regular group.²⁰ We then aggregate the key measures of the three groups, such as the number of booked nights and nightly prices, to the zip code-month level.²¹ Within each group, we rerun the main DID model in Equation (1).

Table 8 reports the results from this three-group analysis, showing the program’s positive and significant impacts on matching performance for all three groups. The results show that the Plus listings experienced the largest increase among the three groups, which is expected given the design of the Plus program. We also find that the close-to-Plus listings enjoyed a positive and significant increase in matching performance, likely because the Plus program nicely distinguishes between the Plus and close-to-Plus groups. Their difference from the Plus listings would be hard to tell without the program, which reduces the search frictions for guests looking for close-to-Plus listings.²² It is noteworthy that the regular listings also demonstrated a considerable increase in the number of booked nights after the program launch.

Our conjecture for why the regular listings see an increased number of booked nights goes beyond the direct impact of reduced search frictions: the improved matching performance because of reduced search frictions may allow hosts of Plus and close-to-Plus listings to raise prices, whereas regular listings may be less likely to change prices. As a result, the regular group may become more attractive with competitive price advantages and, thus, see an increase in booked nights. This reasoning follows the argument from classic quality differentiation literature (e.g., Tirole 1994). When there is more quality differentiation, price competition becomes less intense (but aligns better with quality), and all sellers could see higher revenues (Shaked and Sutton 1982). In our context, the increased booked nights of the Plus and close-to-Plus listings come directly from the reduced search frictions, whereas the increased booked nights of regular listings seem to benefit from these top tiers’ response to the reduced search frictions (i.e., raising prices).

To test this conjecture, we investigate how the Plus program influences the average listing prices of the three groups. Table 9 shows that the Plus program substantially increases the price of the Plus listings (by about \$18 in the full sample and \$19 in the matched sample) and the close-to-Plus listings (by about \$11 in

Table 9. Impacts of the Plus Program on Listing Price of Three Listing Groups

Dependent variable: Average listing price	Plus listings		Close-to-Plus listings		Regular listings	
	(1) Full sample	(2) CEM sample	(3) Full sample	(4) CEM sample	(5) Full sample	(6) CEM sample
AfterPlus	18.039*** (4.166)	19.342*** (4.705)	10.920*** (3.877)	11.606*** (4.360)	4.908 (3.566)	4.462 (3.690)
ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,952	10,045	12,952	10,045	12,952	10,045
Adjusted R ²	0.871	0.865	0.857	0.845	0.898	0.879

Note. Robust standard errors clustered at the zip code level are shown in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

the full sample and \$11.5 in the matched sample). The price increase of the regular group is smaller (about \$5 in the full sample and \$4.5 in the matched sample) and statistically insignificant. This finding is in line with a typical tiering model (Shaked and Sutton 1982) and confirms our conjecture on the competitive price advantages of regular listings. We believe these price increases are due to the Plus program (rather than natural trends) because the parallel-trends assumption holds true for each group, and the change is logically consistent with our other findings.

Taken together, the results in Tables 8 and 9 show that the Plus program concurrently increases the listing prices and the number of booked nights for all types of listings even if these two outcomes are often negatively correlated. In other words, reduced search frictions benefit all listings groups. This is plausible because the Plus program attracts guests who might not book an Airbnb listing otherwise because of high search frictions; the reduced search frictions make the market more attractive and efficient.

The platform is also better off after launching the Plus program for quality differentiation among suppliers. We gather direct evidence by calculating the platform's potential revenue gains (from the contribution of the three groups) using a back-of-the-envelope calculation. For a typical zip code, the total revenue gain ranges from \$17,504 to \$38,163.²³ These results support our conjecture that the market participants (all listing groups and the platform itself) benefit from quality differentiation. This strategy does not marginalize any market participant and is highly rewarding.

4.4. Alternative Explanations

Several alternative explanations of the increased matching performance merit careful study. This section investigates these alternative explanations, and we find little empirical evidence that they are important drivers of the program's impact.

4.4.1. Supply Change in the Market. There is a possibility that the increased matching performance does not result from reduced search frictions but rather is an equilibrium outcome of a decrease in listing supply concurrent with or following the program launch. Because of the introduction of the Plus program, some listings might exit the market because their hosts decide to leave. If the demand holds constant, the number of booked nights for each listing is going to increase. We study the supply change in the market by investigating the program's impact on the listing supply in affected zip codes. Web Table A6 reports the results. Contradictory to the alternative explanation, we do not find any significant change in the listing supply. The increase in booked nights does not seem to come from a decline in the number of listings.

Another concern is, although the Plus program does not change the listing supply, it could influence the number of available nights per listing. The Plus program may motivate hosts to make more nights available on the listing calendar for guests to select from. Web Table A7 reports the results and shows that, although there is a slight increase in available nights associated with the Plus program, it does not undermine the positive impact of the Plus program on matching performance. Additionally, it is possible that the Plus program influences the number and quality of entering/exiting listings, which drives the program's impact. Web Table A8 reports the impacts of the Plus program on entrants and exits and finds no significant patterns.

4.4.2. Change in Listing Quality. Some may wonder if listings receive more booked nights because of a quality enhancement from the Plus program, making them more attractive to guests. In this case, the improved listing quality, rather than reduced search frictions, may drive the matching performance. We test this possibility by observing the change in the listing quality measured by the average review ratings after the program launch. Web Table A9 suggests no significant change in listing quality across all groups of listings after the program launch.

4.4.3. Effects of Airbnb Characteristics. Another concern is whether any listing characteristics would take away the program's impact. Because the program requires hosts/listings to complete a quality check on more than 100 checkpoints, it is possible that the change in some listing characteristics required by the program, rather than the reduced search frictions, boosts matching performance. After examining the requirements of the Plus program, none of the listing characteristics, such as being instantly bookable, business-ready, free cancellation, and being an entire home, are a prerequisite for the Plus program. Yet, to formally address this concern, we control these listing characteristics (aggregated at the market level) to test if the program's effect remains the same. We also include the number of listings and the number of hosts to control the size of the market. Web Table A10 presents the results. Consistent with our main findings, the impact of the Plus program remains positive and significant after controlling Airbnb characteristics. We find no evidence supporting the suspicion that Airbnb characteristics would take away the program's impact.

4.4.4. Alternative Matching Systems. The Plus program, in operation, is an iconic certificate as part of the website design. It is reasonable to wonder whether other matching systems in the website design (filters, sorting tools, host badges, reputation systems, etc.) might play a similar role as the Plus program, therefore confounding

its effect. We argue this is not likely a serious concern because both affected and unaffected markets had these alternative matching systems in place during our study period. In addition, there was no change in these systems during our study period, which is likely to coincide with the program launch. Nevertheless, in the analysis, we still include a plethora of matching systems that could possibly serve a similar role as the Plus program does. These matching systems include popular website designs used by guests to search, filter, or sort listings such as the super host badge, listing prices, and review rating. We measure these alternative systems through the mean and standard deviation of their values.²⁴ Web Table A11 presents the results. The effect of the Plus program remains consistent with our main specification controlling for these alternative matching systems. Alternative matching systems also have positive (and sometimes significant) effects on matching performance. Different from these matching systems, which have been commonly implemented on two-sided platforms, the Plus program adds unique value toward higher matching performance.

4.4.5. Change in Demand. Some may worry that a change in demand may drive our results. To make sure our results are immune to a demand change, we take several steps. First, we investigate the relative time models for two demand proxies: the total number of booked nights and the average number of newly added booking-verified reviews.²⁵ Both measures approximate the potential demand before the program launch and indicate the realized demand. Evidence from relative time models suggests the affected and unaffected markets experience a similar demand trajectory before the program launch (see Web Table A12). It is, therefore, unlikely that the different trends in demand before the program drive our results. Second, we use Google searches for hotels or Airbnb to proxy for the demand for lodging in a market as Barron et al. (2020) did. Higher Google search volumes for hotels or Airbnb in a market may indicate higher demand for lodging in this market. We download the Google search trend data using “city + Airbnb” and “city + hotel” as the search keywords over our study period. The keywords are, for example, “Denver Airbnb” or “Boston hotel.” Using both *t*-tests and nonparametric Wilcoxon (1945) tests, we find no statistical difference in the search volumes before and after the program launch, suggesting no relative change in demand in these markets. Finally, we include these trend data as controls in our main specification and the results hold. We report the analyses in Web Table A13.

4.4.6. An Entry Model. We discuss the application process by hosts to the Plus program in Section 3.1. There may be concerns that Airbnb selected certain markets to launch the Plus program but not others. We apply a

COX proportional hazards model to investigate if any market-level characteristics predict the entry of the Plus program. The results are reported in Web Table A14. None of the market-level characteristics, especially the number of booked nights or the average listing price (our primary dependent variables), are significantly related to the program launch. This finding alleviates the concern of reverse causality and selection bias.

4.5. Robustness Checks

In this section, we perform additional checks to further strengthen the robustness of our findings. Overall, these checks yield consistent results.

4.5.1. Two-Way Fixed-Effects Estimators with Heterogeneous Treatment Effects. The recent literature shows concerns about the accuracy of the treatment effect in a DID model estimated with two-way fixed effects with multiple treatments (or staggered adoption) because it is a weighted average of multiple DID estimators (Goodman-Bacon 2021). In our context, the concern is that the two batches of cities affected by the Plus program may have heterogeneous treatment effects. Furthermore, the effects may be dynamically changing, and the weights of multiple DID estimators may be negative (de Chaisemartin and D’Haultfœuille 2020). We follow de Chaisemartin and D’Haultfœuille (2020) by adopting the two-way fixed-effects estimator with heterogeneous treatment effects (the CD estimator, hereafter) to address these concerns.

We first test whether the weights are negative. We find that our treatment effect is a weighted sum of 148 local average treatment effects, and none of them receives a negative weight. Second, we obtain the CD estimator, which is robust to heterogeneous treatment effects and dynamic effects. The detailed results are reported in Web Table A15. We find the results highly consistent with our original DID estimates on both the full and CEM samples with an even larger treatment effect after accounting for heterogeneous and dynamic treatment effects using the CD estimator.

4.5.2. Alternative Measures of Matching Performance. In the main analyses, we used the number of booked nights in a market as the measure of matching performance. This measure reflects the change in available nights on the host calendar between two consecutive months. Because some hosts may periodically block their calendar (without withdrawing the listing from the platform) for occasions such as home improvement or private use, this measure is likely an upper bound of the actual booked nights. We address this concern by adopting two alternative measures of matching performance. First, we use the average number of booking-verified reviews left by guests for all listings in a month. Because it is unlikely for every guest to leave a review

after staying in a property, this measure is conservative and serves as a lower bound of the actual booked nights. Second, we measure matching performance using the booking rate rather than the absolute booked nights. The booking rate is measured as the ratio of the number of booked nights to the total available nights of all listings in a market. For example, if the number of booked nights is 100 nights and the total number of available nights is 200 nights, the average booking rate would be 0.5. We replicate the specification in Equation (1) using the aforementioned two measures as the dependent variables. Web Table A16 presents the estimates and shows similar results to what we previously found.

Some may also question if the increased average booked nights per listing may not necessarily mean the market is seeing more total transactions. We use both the raw total booked nights and the natural-log transformation of booked nights to proxy for market-level total bookings and find qualitatively consistent results, reported in Web Table A17.

4.5.3. Alternative Measures of Moderators. Our binary measures of discovery and evaluation costs allow us to divide the markets into two groups for comparison. The interaction terms of the dummy variables and the program launch can represent the average treatment effect of the Plus program on matching performance for each group. Some may wonder if continuous moderators are more meaningful. For cross-validation purposes, we report the results using continuous moderators in Web Table A18.

4.5.4. Alternative Clustered Standard Errors. In our main analyses, we cluster the standard errors at the zip code level per our unit of analysis (zip code and year-month). This approach is common in empirical studies utilizing the DID strategy (Callaway and Sant'Anna 2021, Wang and Overby 2021). An alternative option is to cluster at the treatment level, which is the city level in our setting. We replicate all of our main results with city-level clustered standard errors as a conservative estimation and report the results in Web Table A19. Although the standard errors become larger as expected, our primary findings still hold.

4.5.5. Alternative Formats of Dependent Variables. In our main analyses, we use the raw values of booked nights and listing prices as our key dependent variables. Some may wonder if the results hold the same if we instead use the log-transformed formats of the dependent variables. We check the robustness of our results using this alternative format in Web Table A20. The findings remain consistent with our main results.

4.5.6. City-Specific Trends and Seasonality. We control two-way fixed effects at the zip code and year-month levels in the main specification. Some may be concerned that each city might follow a specific trajectory that is different from the others, and the city-specific trends and/or seasonality are not completely captured by the fixed effects. To address this concern, we first construct linear city-specific trends by interacting dichotomous variables, each indicating a city, with a sequential numbering of the time periods beginning with one for the first month in our sample (i.e., for each city, there is a unique time trend variable valuing from 1 to 27). We also construct quadratic city-specific trends by using the square term of these time periods. The purpose is to control for any unobserved linear or quadratic city-level trends that influence Airbnb markets but are different from city to city (Zervas et al. 2021, Chen et al. 2022a). Besides the city-specific trend approach, we further follow a common practice to group months by quarters (e.g., first, second) and control city-specific seasonality in our main analysis. Web Table A21 reports the estimation results. We find no evidence that city-specific trends or seasonality are valid drivers of the program's impacts.

4.5.7. Placebo Test with Random Treatments. A common concern with the DID strategy is that some random events other than the focal event (i.e., the program launch) drive the changes in matching performance. We run a placebo test to show that the effect of the Plus program would not have materialized had we used an artificial program launch date as suggested by Rothstein (2010). Specifically, we use bootstrapped simulations to repeatedly assign random treatments. For each draw, we first randomly treated 28 zip codes in February 2018 as if they were affected by the Plus program (the 28 zip codes are to match the treated zip codes in San Francisco). We further randomly select 118 other zip codes in November 2018 (to simulate the same number of treated zip codes from Denver; Nashville; New Orleans; and Washington, DC). We then replicate our main regressions from Equation (1). We use the number of both booked nights and booking-verified reviews (an alternative measure) to measure matching performance. Web Figure A3 illustrates the distributions of the estimated coefficients from simulations with 1,000 draws for the two measures of matching performance. The distribution is centered near zero. The results of this placebo test further support our finding that the Plus program drives the increase in matching performance rather than other random events.

5. Concluding Remarks

Matching makes or breaks a P2P platform. As platform-based P2P markets continue to proliferate and

scale, so do discovery and evaluation costs. These search frictions are often regarded as a primary hurdle to matching, which is why P2P platforms are actively combatting them. We conduct timely research to study the role of Airbnb Plus—a program differentiating listings based on quality—in reducing search frictions and how the effects vary among different product tiers. These problems are under-addressed in the literature yet critical to P2P platform design and supplier management. The Plus program was rolled out in several U.S. cities over time, providing an empirical opportunity to examine the effect of quality differentiation on matching performance among markets.

Being a costly program, the Plus program requires in-person, off-line inspections of the listing against more than 100 checkpoints. Such a quality endorsement by platforms addresses search frictions that widely exist in experience goods such as accommodation properties. It is clear that Airbnb, a once-grassroots service and cheaper alternative to hotels, aspires to become a mainstream hospitality company. We find the Plus program is welfare-rewarding to all market participants. At the market level, quality differentiation is effective in facilitating matching on P2P platforms by reducing search frictions (toward improved matching performance). Overall, the platform experiences an 85.04- to 98.60-night increase in booked nights in a typical Airbnb market. The impact is especially salient in markets with higher discovery costs and evaluation costs (collectively, larger search frictions), that is, markets with a higher density of listing options and a more heterogeneous listing supply. At the group/tier level, the Plus (and close-to-Plus) listings experience the premium of increased booked nights along with higher pricing because of reduced search frictions, whereas regular listings benefit from increased booked nights because of competitive prices compared with the other two groups.

Our study, among the first to comprehensively evaluate a costly quality differentiation program initiated by the platform, provides new insights into the literature regarding search theory and matching, search frictions, platform design, and P2P markets. Our findings also inform P2P platforms of the importance of supplier management (more specifically, differentiating suppliers by quality) as their offerings continue to grow and diversify. Such a differentiation strategy profoundly improves matching performance. In addition, we find that all types of listing groups (i.e., the Plus, close-to-Plus, and regular listings) benefit from the Plus program. This finding indicates that quality differentiation benefits the entire market and calls for further research into the optimal number of quality tiers. Given that one prerequisite for quality differentiation to function as intended is an obvious quality

signal, more tiers are not necessarily better. Finally, our study informs the ongoing innovations across P2P platforms, which have begun to experiment with various platform design strategies to improve matching performance.

Acknowledgments

The authors express their sincere appreciation to the department editor, Hemant Bhargava, the associate editor, and three anonymous reviewers. Their invaluable feedback has significantly enhanced the quality of the work. For helpful comments, the authors are grateful to the seminar attendees at University of California, Davis, the University of Georgia, and Southern Methodist University, as well as anonymous reviewers and participants at the Conference on Information Systems and Technology 2020. All authors contributed equally to this work.

Endnotes

¹ See <https://www.uber.com/us/en/ride/uberlux/>, retrieved in May 2022.

² See <https://www.airbnb.com/plus>, retrieved in May 2022.

³ See <https://news.airbnb.com/about-us/>, retrieved in January 2023.

⁴ See <https://news.airbnb.com/airbnb-unveils-roadmap-to-bring-magical-travel-to-everyone/>, retrieved in May 2022.

⁵ There are five U.S. cities that experienced the program launch during our study period. They are San Francisco (February 2018); Denver; Nashville; New Orleans; and Washington, DC (November 2018). See <https://airbnbuniverse.com/airbnb-plus-cities/>.

⁶ Because of data unavailability prior to the program launch, there are other U.S. cities that experienced the Plus program but were not included in our sample, such as Austin, Chicago, and Los Angeles.

⁷ More details about the Plus program and its requirements can be found at <https://www.airbnb.com/help/article/2195>, retrieved in March 2023.

⁸ In general, there are four types of homes on Airbnb: entire home, private room, shared room, and hotel rooms. Different from private rooms and shared rooms in which hosts and guests share the space, entire homes are a preferred choice for guests who appreciate hotel-like private spaces without hosts or other guests' presence. See <https://www.airbnb.com/help/article/5/what-does-the-room-type-of-a-listing-mean>, retrieved in May 2022.

⁹ This test focuses on whether the Plus program is able to screen higher quality listings from non-Plus listings without any assumption on the impact of the Plus program on listing quality.

¹⁰ Super hosts are experienced hosts who provide a shining example for other hosts and extraordinary experiences for their guests. To become a super host, hosts need to have an account in good standing and meet requirements such as completing at least 10 trips, maintaining a 1% percent cancellation rate (1 cancellation per 100 reservations) or lower, and maintaining a 4.8 overall rating (equaling to 96/100), etc. See <https://www.airbnb.com/help/article/829/how-do-i-become-a-superhost>, retrieved in May 2022.

¹¹ As a robustness check, we also measure matching performance using the number of booking-verified reviews and booking rates. The reasoning and results of this analysis are reported in Section 4.5.

¹² Because the treatment (i.e., the program launch) is zip code and time specific, $AfterPlus_{it}$ effectively represents $Treated_i + After_t + Treated_i \times After_t$ in classic two-period DID models in which $Treated_i$ is absorbed by the zip code fixed effects and $After_t$ is absorbed by the time fixed effects.

¹³ Specifically, the CEM approach involves two separate matching tasks corresponding to the staggered launch of the Plus program by city batch. In the first task, we match all zip codes in San Francisco (i.e., affected by the Plus program in February 2018) with all of the zip codes that were not affected at that time. In the second task, we match all zip codes in Denver; Nashville; New Orleans; and Washington, DC (affected by the Plus program in November 2018) with the remaining unaffected zip codes. In this process, the CEM approach slightly reduces the number of observations but largely mitigates the imbalance in covariates between affected zip codes and unaffected zip codes. CEM weights are used to account for the different numbers of affected and unaffected zip codes.

¹⁴ This matching approach allows varying numbers of treated and control units and is preferred to the $k2k$ matching approach because it uses maximal information and maintains as many matched zip codes as possible (Blackwell et al. 2009).

¹⁵ For unaffected zip codes, we consider the launch time of the matched affected zip codes as the treatment time. When an unaffected zip code is not matched to any affected zip code, the second batch treatment time is considered the treatment time.

¹⁶ Market thickness is considered more of a long-lasting feature in each zip code than a floating indicator because it is determined by the inherent supply of and demand for rental homes in each zip code. We have empirically verified that the program launch did not change the market thickness or listing supply (see Web Table A6), and the classification of thicker/thinner markets is very robust across time (more than 95% of zip codes belong to the same group all the time).

¹⁷ As robustness checks, we also estimate the moderating effect of market thickness using (i) the market thickness dummy, *Higher Market Thickness*, as the moderator to interact with the program dummy; (ii) subsample analyses with two separate samples of markets of higher (above sample median) thickness and lower (below sample median) thickness; and (iii) a continuous measure of market thickness as the moderator to interact with the program dummy. All analyses return qualitatively consistent results resembling the ones from Equation (3). The results are reported in Web Tables A2, A3, and A18.

¹⁸ We do not interact the levels of market thickness with zip code fixed effects because they suffer from perfect collinearity, that is, the level of market thickness is fixed for each zip code.

¹⁹ Similar to the robustness checks of Equation (3), we also estimate the moderating effect of rating dispersion using (i) the rating dispersion dummy, *Higher rating dispersion*, as a moderator to interact with the program dummy; (ii) subsample analyses using two separate samples of markets with higher (above median) rating dispersion and markets with lower (below median) rating dispersion; and (iii) a continuous measure of rating dispersion as the moderator to interact with the program dummy. All analyses return qualitatively consistent results resembling the ones from Equation (4). The results are reported in Web Tables A2, A3, and A18.

²⁰ We check the comparability between Plus and close-to-Plus listings in Web Table A4. More detailed descriptive statistics of the three groups are available in Web Table A5.

²¹ Note that in the unaffected zip codes, all the Plus-related measures are the same as close-to-Plus measures because the close-to-Plus listings from unaffected zip codes also serve as the control group for Plus listings in affected zip codes.

²² Despite no intention to compare the magnitudes of the program's impacts on these two groups, we have also tested the difference between the two coefficients of the Plus and close-to-Plus groups. In the full sample analysis, the p -value of the difference is 0.078; in the CEM sample analysis, the p -value of the difference is 0.095. The coefficient comparison suggests a possible larger impact of the Plus program on the Plus group, evident in a slightly larger increase in the booked nights after the program launch.

²³ We rely on the summary statistics of the full sample to calculate potential revenue gains. The formula is (base price + increased price) $\times$ (base booked nights + increased booked nights) – base price $\times$ base booked nights. “Base” price/booked nights refers to the values before the program launch, whereas “increased” price/booked nights refers to the treatment effects reported in Tables 8 and 9. For a Plus listing, the potential gain is \$238.52 (i.e., $(\$197 + \$18) \times (2.74 + 0.88) - \197×2.74). For a close-to-Plus listing, the potential gain is \$139.35 (i.e., $(\$195 + \$11) \times (2.93 + 0.52) - \195×2.93). For a regular listing, the potential gain is \$109.40 (i.e., $(\$153 + \$5) \times (2.92 + 0.60) - \153×2.92). Using \$109.40 and \$238.52 as lower and upper bounds, for a typical zip code with 160 listings (the sample mean in our data), the total revenue gains range from \$17,504 to \$38,163.

²⁴ Only when there is enough variation can guests utilize these systems to discover potential options. It is, thus, important to observe these systems through both an average measure and a standard deviation measure. Yet we cannot identify the causal impact of any matching systems because of the lack of exogenous shock. All the coefficients mainly reflect their correlations with matching performance.

²⁵ On Airbnb, only guests who have made the trip and stayed at the booked listing can post reviews.

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